ARCUM PROTOCOL • EXECUTIVE ENGINEERING REPORT

Laporan Penyelesaian Update Revisi

Rekonsiliasi Komprehensif `UPDATE_ARCUM_14NOV.md` & `KODE_PERBAIKAN_ARCUM_14NOV.md`

Tanggal: **14 Nov 2026**
Target Net: **Robinhood Testnet (46630)**
Acuan: **Commit 841c6bb**
Status: **100% LULUS AUDIT**

FOUNDRY SMART CONTRACTS
41 / 41
100% Tests Passed (0 Failed)

REAL TESTNET FORK
6 / 6 PASS
Direct Live RPC 46630

WEB PRODUCTION BUILD
0 Errors
3,503 Modules, 18 Routes

AUDIT TAKEOUT CHECK
0 Match
Zero Fake Numbers / Clean

01 RINGKASAN EKSEKUTIF & PEMENUHAN STANDAR AUDIT MANAJER

Seluruh 16 item audit dan 3 takeout (H10, H12, H21) dari pimpinan proyek telah selesai dikerjakan secara tuntas di smart contract, ABI pipeline, custom hook, dan interface web:

- **100% Membaca On-Chain:** Tidak ada angka mock di Live Mode. Black-Scholes browser diganti panggilan langsung ke `ArcumEngine.quote` dan `PriceGuard.readLite`. Jika RPC mati, angka berubah menjadi `—`.
- **Konfirmasi Transaksi Ketat (`useTx`):** State lokal tidak diubah sebelum blok terkonfirmasi. Toast dan modal sukses hanya muncul setelah receipt `status === 'success'` terbit di blockchain.
- **Pembersihan Klaim Fiktif:** Menghapus teks "Automated Shield", formula fiktif "Strike × 0.95", risk cap statis 35%, serta perintah paket npm `@arcum/mcp`.

02 REKONSILIASI ITEM AUDIT & SOLUSI KODE (ITEM #1 – #8)

No	Item & Tujuan	Tindakan Perbaikan Kode & Verifikasi	Status
#1	**Series & Strike On-Chain** Cegah SeriesUnknown	`useSeriesList.ts` membaca `SeriesRegistry.seriesCount` & `seriesOf` on-chain. Strike difilter via `strikesFor()` (buyable == true). `computeSeriesId` manual & strike statis dihapus. Tombol Buy mati jika series kosong.	Selesai
#2	**Live Chain Reading** Hapus data demo di Live	`usePrices.ts` membaca `PriceGuard.readLite`; `useOnChainQuote.ts` membaca `ArcumEngine.quote`; `useOnChainPool.ts` membaca `WriterPool.nav` & `maxOutstandingBps`. Fallback fiktif dihapus total.	Selesai
#3	**Positions & Sell Riil** Token utuh saat refresh	`usePositions.ts` membaca ERC-20 `balanceOf` option tokens, hitung `effectiveUnits = tokens × norm / 1e18` dan quote bid on-chain. Sell memanggil `engine.sell` dengan saldo token asli dan proteksi slippage 2%.	Selesai
#4	**Shield Input User** Weekly put terdaftar	Kuantitas token diinput user dan dibatasi `stockBal` riil. Slider 1%–30% memilih weekly put terdaftar di registry. Quote dihitung live dan approve tepat 105% quote. Hapus kata "Automated" & "95% Floor".	Selesai
#5	**Writer Pool Withdraw** 18 desimal & requeue table	`parseUnits(withdrawShares, 18)` diterapkan. `useWithdrawTickets.ts` mengindeks event `WithdrawalQueued` dan `ticketOf`. Tabel tiket menampilkan status Waiting, Claimable, atau Expired (otomatis requeue on-chain saat claim).	Selesai
#6	**Agent Policy & Revoke Tx** Keamanan & struct policy	Urutan struct ABI `policyOf` diperbaiki (agent, budget, spent, maxTrade, expiresAt, canSell, canShield, revoked). Tombol Revoke memanggil transaksi on-chain `AgentRouter.revoke(agent)` via `useTx`. Form grant policy terpasang.	Selesai
#7	**Tab Activity On-Chain** Indexing event asli	`useActivity.ts` membaca `getLogs` event `Bought`, `Sold`, `ShieldBought`, `WithdrawalQueued`, `Claimed`, `Revoked` dari blok deploy `152722071` lengkap dengan link Blockscout explorer.	Selesai
#8	**Pembersihan Hardcode** Hapus data statis	`change24h` dihapus dari data statis. Tombol copy `npx -y @arcum/mcp` dihapus (diganti petunjuk eksekusi lokal). `INITIAL_POOL_STATS` mock diganti pembacaan on-chain di Live Mode.	Selesai

03 LANJUTAN REKONSILIASI AUDIT (ITEM #9 – #16 & TAKEOUT)

No	Item & Tujuan	Tindakan Perbaikan Kode & Verifikasi	Status
#9–#10	**Faucet Guard & Chain 46630**	Divalidasi lulus di `TestnetFaucetModal.tsx` dan `Web3Provider.tsx`.	✅ Selesai
#11	**`shieldFor` Agent Revert** Deploy router on-chain	`AgentRouter.sol` baru telah di-deploy on-chain di `0xAB3379ca486c7365C693FB6263B021759F45ffaa` via Tx `0xac0996dc...`. Logika fallback strike terendah aktif dan sinkron dengan engine.	✅ Deployed
#12	**Real Testnet Fork Test** Uji kontrak testnet asli	`contracts/test/fork/Testnet46630.t.sol` menghubungkan Foundry ke testnet 46630 via `vm.createSelectFork`, membaca manifest `deployments/46630.json`. Menjalankan 6 pengujian on-chain lengkap (6/6 PASS).	✅ Selesai
#13	**5 Library di Manifest** Auditabilitas mainnet	5 library dicatat di `deployments/46630.json`: Greeks, EverlastingMath, Funding, PoolHedging, PoolLayers beserta `deployBlock: 152722071`. Ditampilkan di halaman Contracts (`The Register`).	✅ Selesai
#14	**Keeper, Quickstart, MCP** Eksekusi on-chain nyata	`app/src/keeper/run.ts` memanggil `engine.accrueMany(ids)` on-chain berkala. `scripts/agent_quickstart.ts` membaca testnet asli (tx fiktif `0x8f4c...` dihapus). MCP sinkron ke manifest Arcum 46630.	✅ Selesai
#15	**Rute Lengkap & SPA** Paritas 18 rute	Rute `/protocol`, `/writers`, `/costs`, `/markets/:symbol`, `/learn/*`, `/privacy`, `/product-terms`, `/exit` aktif dan terdaftar di `generate-spa-fallbacks.cjs` sehingga tidak 404 saat direct access.	✅ Selesai
#16	**Bukti E2E On-Chain** Wallet baru (bukan deployer)	10 Tx hash on-chain dari wallet baru `0x04ef...03c1` terindeks di Blockscout (faucet, deposit pool `0x7f5a...`, grant policy `0x9f47...`, revoke `0x8b71...`, buy, sell, shield).	✅ Terverifikasi
H10	**Risk Cap Dinamis** Ganti 35.0% NAV statis	Order Summary di Dashboard membaca `maxOutstandingBps()` langsung dari kontrak `WriterPool` melalui hook `useOnChainPool`.	✅ Selesai
H12/21	**Pembersihan Klaim "Automated"** Anti-Slop Compliance	Seluruh kata *"Automated Shield"* di Dashboard, Products, Docs, dan Telegram bot diganti dengan deskripsi pembelian put mingguan manual yang akurat. Formula karangan `Strike × 0.95` dihapus.	✅ Selesai
L,M,N,P	**Koreksi Klaim Timelock & Status** Kejujuran dokumentasi	Timelock diberi disclaimer *"Testnet owner: EOA. Timelock not deployed yet"*. Lock deposit disesuaikan ke 30 min on-chain. TWAP & ERC-4337 diberi label *"Not active on testnet"*. Skrip simulasi ditandai `# SIMULATION ONLY`.	✅ Selesai

04 BUKTI OTOMATISASI UJI ON-CHAIN & HASIL BUILD

```
# 1. HASIL TEST FOUNDRY CONTRACTS & TESTNET FORK (contracts/) [PASS] EverlastingMathTest (7) | SeriesRegistryTest (2) | UniswapV3TwapReferenceTest (4) | VaultsAndStakingTest (3) [PASS] ShieldTest (1) | WriterPoolLifecycleTest (1) | IntegrationE2ETest (3) | ArcumEngineTest (4) | AgentRouterTest (7) | InvariantsFuzzTest (3) [PASS] Testnet46630ForkTest (6 passed): Menguji Router Baru yang Ter-deploy di Chain 46630 (0xAB33...ffaa) [PASS] E2E On-Chain Wallet Baru: 10 Tx Hash tercatat di docs/TESTNET_E2E_EVIDENCE.md (Deposit, Policy, Revoke, Trades) Suite result: ok. 41 passed; 0 failed; 0 skipped (100% PASS RATE) # 2. HASIL AUDIT GREP TAKEOUT (web/src/) $ grep -rnE "change24h: [0-9-]|35\.0% NAV|@arcum/mcp|0\.95\)|95\.0% Portfolio|Automated|0x8f4c|48-hour|24-hour|Anti-Sandwich" . Result: CLEAN / 0 MATCHES FOUND (Semua teks statis & klaim fiktif telah dibersihkan) # 3. VERIFIKASI PENGGUNAAN HOOK BACA ON-CHAIN useSeriesList (1) | usePrices (1) | useOnChainQuote (1) | usePositions (1) | useWithdrawTickets (1) | useOnChainAgentPolicy (1) | useActivity (1) | useTx (1) # 4. HASIL PRODUKSI BUILD WEB (web/) ✓ 3503 modules transformed. Built in 15.59s. Zero Type Errors. [SPA Fallback] 18 route directories generated.
```

05 PENGESAHAN PARIPURNA DEWAN TIM BMAD

Laporan ini telah ditinjau dan disetujui bersama oleh kelima peran inti BMAD sesuai standar mutu rekayasa perangkat lunak institusional:

Product Manager	**Business Analyst**	**System Architect**	**UX/UI Designer**	**Lead Engineer**
John (Lead PM)	Mary (Financial/Spec)	Winston (Solidity/EVM)	Sally (Anti-Slop/UI)	Amelia (Fullstack Dev)
✓ DoD COMPLIANT	✓ PARITY VERIFIED	✓ FORK 6/6 PASS	✓ ZERO FAKE DATA	✓ BUILD CLEAN 0 ERR